Differential and Integral Calculus

Chapter 3 — Derivatives, Integrals, and Series

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3.1 The Derivative as a Limit

Given a function f defined on an open interval containing a point x, the derivative $f'(x)$ is defined as the limit of the difference quotient: $f'(x) = \lim_{h \to 0} [f(x+h) - f(x)] / h$, provided this limit exists. Geometrically, the derivative represents the slope of the tangent line to the graph of f at the point $(x, f(x))$.

From the definition one can derive the power rule, $d/dx\, [x^n] = n x^{n-1}$, as well as the rules for sums, products, and quotients. The chain rule, $d/dx\, [f(g(x))] = f'(g(x))\, g'(x)$, allows us to differentiate compositions. Applying these rules, one finds for example that $d/dx\, [\sin(x)] = \cos(x)$ and $d/dx\, [\exp(x)] = \exp(x)$, with the latter being the unique non-trivial function that equals its own derivative.

The function $f(x) = x^2$ has derivative $f'(x) = 2x$, vanishing at the origin. Its graph is a parabola with vertex at $(0,0)$ opening upward; the tangent at any point (a, a^2) has equation $y = 2 a x - a^2$. Plotting the family of tangents traces out the parabola itself as their envelope.

3.2 The Definite Integral and the Fundamental Theorem

The definite integral of a continuous function f over an interval $[a, b]$ is defined as the limit of Riemann sums: $\int_a^b f(x)\, dx = \lim_{n \to \infty} \sum_{k=1}^n f(x_k)\, \Delta x$. Geometrically, this represents the signed area between the graph of f and the x-axis over $[a, b]$.

The Fundamental Theorem of Calculus links differentiation and integration. Part I states that if $F(x) = \int_a^x f(t)\, dt$, then $F'(x) = f(x)$. Part II states that $\int_a^b f(x)\, dx = F(b) - F(a)$, where F is any antiderivative of f. Together they reduce many integrals to algebraic computations.

Consider the area under the standard normal density $\phi(x) = (1/\sqrt{2\pi}) \exp(-x^2/2)$. Its integral over the entire real line equals 1, but no elementary antiderivative exists; the cumulative distribution function $\Phi(x)$ must be evaluated numerically. Plotting $\phi(x)$ yields the iconic bell curve centered at zero with inflection points at $x = \pm 1$.

3.3 Taylor Series and Approximation

If f is infinitely differentiable at a point a, its Taylor series around a is $f(x) = \sum_{n=0}^\infty [f^{(n)}(a) / n!] (x - a)^n$. When this series converges to $f(x)$ on a neighborhood of a, the function is called analytic. The Maclaurin series, the special case $a = 0$, gives the familiar expansions $\exp(x) = \sum x^n / n!$, $\sin(x) = \sum (-1)^n x^{2n+1} / (2n+1)!$, and $\cos(x) = \sum (-1)^n x^{2n} / (2n)!$.

Truncating the series at degree N produces a polynomial approximation whose error is governed by the Lagrange remainder $R_N(x) = f^{(N+1)}(c) (x - a)^{(N+1)} / (N+1)!$ for some c between a and x. As N grows, the polynomial graph hugs f more and more tightly within the radius of convergence, while diverging outside it.

Plotting the partial sums of the Maclaurin series for $\sin(x)$ reveals an instructive pattern: each successive odd-degree polynomial closely tracks $\sin(x)$ over an ever-widening interval before peeling away. This visual demonstrates both the local accuracy of Taylor approximation and the inherent limitation of any finite truncation.